AYE
TECH HUB

• MEP ENGINEERING • LESSON 06

Cooling Systems and Chillers

How Buildings Are Cooled — From Split AC to Central Chiller Plants

Lesson 06 of 30 | MEP Engineering Programme | AYE Tech Hub

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- The vapour compression refrigeration cycle — how all mechanical cooling works
- Chiller types: air-cooled, water-cooled, absorption — selection criteria
- COP, EER and IPLV — measuring and comparing chiller efficiency
- Chilled water system design: primary-secondary-tertiary pump loops
- Cooling tower types, Legionella risk management, and water treatment
- Fan coil units (FCUs) and air handling units (AHUs) — selection and sizing
- Complete chiller plant selection example for a 5-storey office building

The Vapour Compression Refrigeration Cycle

All mechanical cooling systems — from a domestic refrigerator to a 10,000 kW industrial chiller — operate on the same fundamental principle: the **vapour compression refrigeration cycle**. Understanding this cycle is the prerequisite for designing, specifying, and commissioning any cooling plant in a building or industrial facility. The cycle exploits the fact that liquids absorb heat when they evaporate and reject heat when they condense — the same physical phenomenon that makes sweating cool your body.

Vapour Compression Refrigeration Cycle — The Heart of All Mechanical Cooling

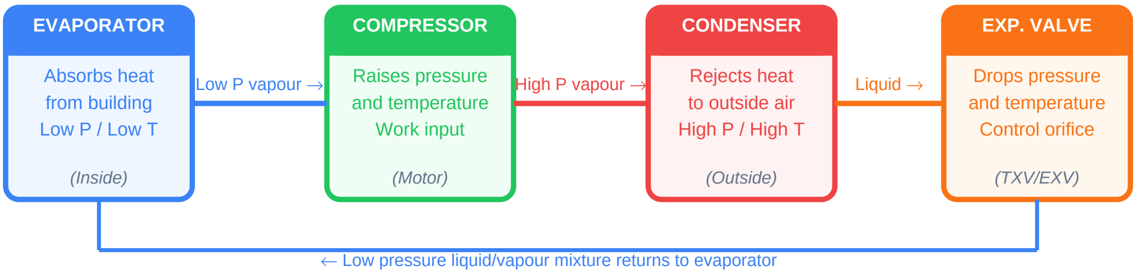

Fig 1 — Refrigerant circulates continuously: absorbs heat at evaporator → compressor raises pressure → condenser rejects heat → expansion valve drops pressure

The Four Stages of the Cycle

Stage	Component	What Happens to Refrigerant	Heat Transfer Direction
1 — Evaporation	Evaporator coil (inside building)	Low-pressure liquid refrigerant absorbs heat from building air/water and evaporates to a low-pressure vapour. Temperature: typically –5 to +8°C	Heat flows FROM building INTO refrigerant (useful cooling effect)
2 — Compression	Compressor (motor-driven)	Low-pressure vapour is compressed to high pressure, raising its temperature to 60–80°C. This is where electrical energy is consumed	No heat transfer — work input raises temperature and pressure
3 — Condensation	Condenser (outside building or cooling tower)	High-pressure, high-temperature vapour rejects heat to outdoor air or cooling water and condenses to high-pressure liquid. Temperature: 35–50°C	Heat flows FROM refrigerant TO outdoor air or cooling water
4 — Expansion	Expansion valve (TXV or EXV)	High-pressure liquid passes through a restriction — pressure drops rapidly, cooling the refrigerant to evaporator temperature. No work input or output	No net heat transfer — pressure/temperature drop is isenthalpic

Refrigerants Used in Modern Cooling Systems

The refrigerant is the working fluid that carries heat around the cycle. Refrigerant selection significantly affects system efficiency, safety, environmental impact, and compliance with international regulations. The Kigali Amendment (2016) mandates phasing down high-GWP HFC refrigerants globally — engineers must specify compliant refrigerants in all new designs:

Refrigerant	Type	GWP	ODP	Status	Typical Application
R-410A	HFC blend	2,088	0	Phase-down — avoid in new designs	Legacy residential and commercial AC, split systems
R-32	HFC (single component)	675	0	Transitional — widely used 2020–2030	Modern split AC, VRF systems, heat pumps
R-134a	HFC	1,430	0	Phase-down in most sectors	Chillers, automotive AC, medium-temp refrigeration
R-407C	HFC blend	1,774	0	Phase-down — legacy systems only	Retrofit of R-22 systems, commercial chillers
R-1234yf	HFO (low-GWP)	4	0	Preferred — growing adoption	Automotive AC, chillers, new residential systems
R-1234ze	HFO (low-GWP)	7	0	Preferred — large chillers	Centrifugal and screw chillers, data centre cooling
R-290 (Propane)	Natural HC	3	0	Preferred where safety permits	Small residential AC, commercial refrigeration
R-717 (Ammonia)	Natural	0	0	Best efficiency — industrial only	Industrial refrigeration, large process cooling

Chiller Types and Selection Criteria

A **chiller** is a complete refrigeration machine that produces chilled water (typically at 6–8°C) for distribution to air handling units, fan coil units, and other cooling terminals throughout a building. The chiller is the central energy-consuming component of a central cooling plant — selecting the right type, size, and efficiency rating has a major impact on the building's lifetime energy cost and carbon footprint.

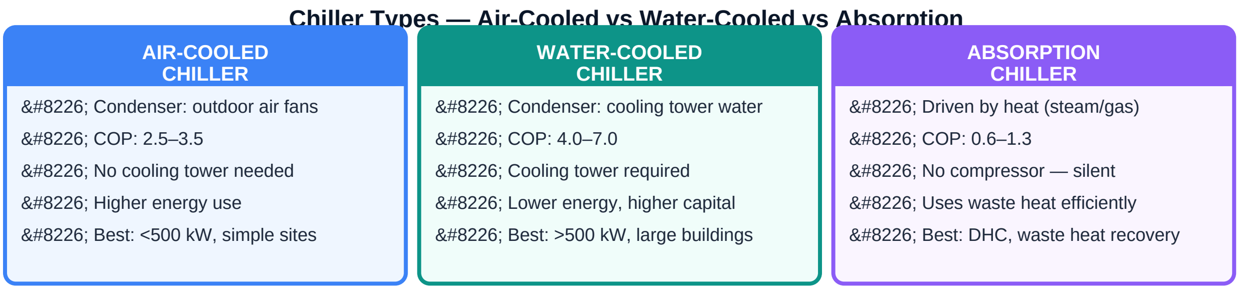

Fig 2 — Water-cooled chillers are most efficient for large buildings; air-cooled for simplicity; absorption for waste-heat recovery

Selection Factor	Air-Cooled Chiller	Water-Cooled Chiller	Absorption Chiller
Cooling capacity range	20 kW to 1,500 kW	200 kW to 50,000+ kW	100 kW to 20,000 kW
COP (full load)	2.5 to 3.8	4.5 to 7.5 (centrifugal: up to 9+)	0.5–0.8 (single effect), 1.0–1.3 (double)

IPLV (part load)	3.0 to 4.5	5.0 to 9.0+	N/A — minimal part-load variation
Water consumption	None — air-cooled	Cooling tower: ~2–4 L/kWh rejected	None (for absorption with waste heat)
Space requirement	Outdoor unit required (rooftop/ground)	Machine room + cooling tower space	Larger footprint than compressor chillers
Energy source	Electricity only	Electricity (compressor)	Heat: steam, hot water, gas, waste heat
Maintenance complexity	Low — no cooling tower	Medium — cooling tower water treatment critical	High — corrosion, crystallisation risk in tubes
Best application	Buildings <500 kW, limited water	Large buildings, district cooling, central plants	Buildings with waste heat or low-cost thermal energy
Approximate installed cost	$400–800 per kW	$250–600 per kW + tower	$600–1,200 per kW

Chiller Efficiency Metrics — What the Numbers Mean

Metric	Formula	What It Measures	Typical Good Value
COP (full load)	Cooling output (kW) / Power input (kW)	Efficiency at 100% load — used for energy cost calculations at design conditions	Air-cooled: 3.0+ | Water-cooled: 5.5+
EER	Cooling output (BTU/h) / Power input (W)	Same as COP but in imperial units. EER = COP × 3.412	Air-cooled: 10+ BTU/Wh | WC: 18+
IPLV / NPLV	Weighted average of COP at 100%, 75%, 50%, 25% load	Part-load efficiency — most important for real buildings that rarely run at 100%	Air-cooled: 4.0+ | Water-cooled: 6.0+
kW/TR	Power input (kW) / Cooling capacity (Tons of Refrigeration)	Energy consumption per ton — lower is better	Air-cooled: <1.0 kW/TR | WC: <0.55 kW/TR

ASHRAE 90.1 minimum efficiency requirements must be met for all new chiller installations. Many green building standards (LEED, BREEAM) require chillers significantly above these minimums. Always specify IPLV rather than full-load COP — buildings spend less than 2% of annual hours at 100% load; part-load performance determines real-world energy consumption.

Chilled Water System Design

In most medium-to-large commercial buildings, the chiller produces chilled water that is distributed through a piping network to cooling terminals. The design of this distribution system — pump arrangement, pipe sizing, control strategy — is just as important as the chiller selection itself. Poor distribution design can negate the efficiency gains from a high-efficiency chiller.

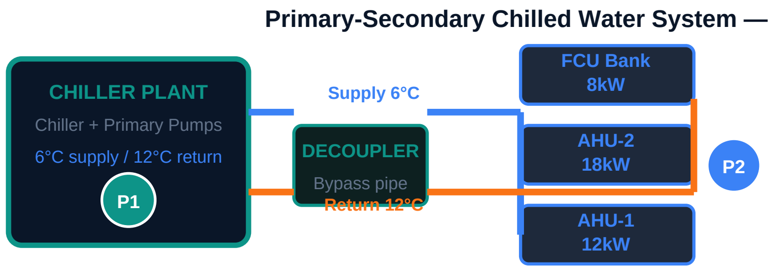

Fig 3 — Primary loop (fixed flow through chiller) decoupled from secondary loop (variable flow to loads) for energy efficiency

Chilled Water System Configurations

Configuration	Description	Best Application	Advantage
Primary only (constant flow)	Single pump circuit; chiller and all loads in one loop at constant flow rate	Small systems <500 kW; simple buildings	Simple, low capital cost; easy to commission and maintain
Primary-secondary (decoupled)	Separate primary loop (through chiller, constant flow) decoupled from secondary loop (to loads, variable flow) via a bridge pipe	Medium systems 500–2000 kW; multiple zones	Energy saving via variable secondary pumps; chiller operates at stable flow
Primary variable flow (PVF)	Single loop; flow through chiller varies with demand. Requires minimum flow bypass at chiller	Modern efficient large plants	Maximum pump energy saving; fewer pumps; complex controls
Primary-secondary-tertiary	Three pump levels: plant, distribution, and individual building or zone level	District cooling; campus plants serving multiple buildings	Maximum flexibility for campus-scale distribution; each building controlled independently

Chilled Water Design Parameters

Parameter	Standard Value	High-Efficiency Value	Impact of Deviation
Supply water temperature	6–7°C (43–45°F)	8–10°C (high-delta T)	Higher supply temp → better chiller COP but larger coils needed
Return water temperature	12°C (standard ΔT = 6°C)	14–16°C (high ΔT system)	Higher ΔT → smaller pipes, less pump energy, fewer pumps
Delta T (ΔT = return minus supply)	6°C (standard)	8–10°C (high efficiency)	Low ΔT syndrome: pumps oversized, chiller cycling — very common problem
Chilled water velocity in pipes	1.0–2.5 m/s (main)	1.5 m/s (balanced)	Too fast → noise and erosion; too slow → stratification, poor heat transfer

Pump head (primary)	15–25 m (plant internal)	As low as possible	Higher head → more pump energy; use variable speed drives on secondary
Glycol concentration	0% (above-freezing climates)	25–30% (risk of freezing)	Glycol reduces heat transfer capacity by 5–10%; reduce chiller capacity accordingly

Cooling Towers — Heat Rejection for Water-Cooled Chillers

A **cooling tower** is the heat rejection device for a water-cooled chiller system. It removes the heat absorbed from the building (by the chiller evaporator) plus the heat added by the compressor work, rejecting both to the outdoor atmosphere via evaporative cooling. A well-designed cooling tower system is essential for achieving the high COP values of water-cooled chillers — but it introduces water consumption and a biological risk that must be managed.

Cooling Tower Type	Airflow Method	Water Circuit	Advantage	Limitation
Open circuit — induced draft	Fan pulls air through warm water spray from above	Recirculating — water contacts air directly	Most common; lowest capital cost; highly effective	Legionella risk; water quality management critical
Open circuit — forced draft	Fan pushes air from bottom through falling water	Recirculating — water contacts air directly	Less drift; fan accessible for maintenance	More susceptible to air recirculation
Closed circuit (fluid cooler)	Fan over a coil through which process fluid flows; water sprays over coil	Process fluid never contacts air — separate spray water circuit	Process fluid stays clean; lower fouling risk	Higher capital cost; lower efficiency than open tower
Dry cooler (no evaporation)	Air-cooled finned coil — no water evaporation	Closed — no water evaporation	No water consumption; no Legionella risk; minimal maintenance	Limited to ambient +10°C — poor efficiency in hot weather

Legionella Risk Management — Non-Negotiable Safety Requirement

Legionella pneumophila is a bacterium that thrives in warm water (25–45°C) and can cause Legionnaires' disease — a severe, potentially fatal pneumonia — when inhaled as fine water droplets (aerosols). Open cooling towers create the exact conditions for Legionella growth and produce the aerosol drift that carries the bacteria. This makes Legionella risk management a legal requirement in most countries, not just a best practice.

Control Measure	Target / Standard	Frequency	Responsibility
Water temperature maintenance	Avoid 25–45°C in storage; return water < 35°C at tower inlet	Continuous monitoring	BMS / controls engineer

Biocide dosing (oxidising)	Free chlorine 0.5–1.0 mg/L OR bromine equivalent	Continuous dosing; verify 3× per week	Water treatment specialist
Biocide dosing (non-oxidising)	Rotate quarterly to prevent resistance	Monthly or quarterly	Water treatment specialist
Blowdown / bleed-off	Maintain conductivity < 2,500 µS/cm (or per water quality plan)	Automatic bleed valve; weekly check	Maintenance engineer
Physical cleaning (tower fill)	No visible biofilm or scale on fill media	Monthly inspection; biannual cleaning	Specialist cleaning contractor
Legionella water sampling	Target: no growth at 100 CFU/L; action at 1,000 CFU/L; shutdown at 10,000 CFU/L	Quarterly minimum; monthly on risk sites	Accredited laboratory
Risk assessment review	Written scheme of control per HSE L8 / CIBSE TM13	Annual review; after any system change	Responsible person (owner/operator)

LEGAL REQUIREMENT — In the UK (HSE ACoP L8), EU, and most MEP jurisdictions, building owners must maintain a written Legionella risk assessment and a scheme of control for all cooling towers. Failure to comply following a Legionnaires' outbreak can result in criminal prosecution of building owners and facilities managers. Always specify cooling tower water treatment in the MEP specification, and ensure the client's FM team has a maintenance contract with a water treatment specialist before handover.

Cooling Terminals — FCUs, AHUs and Fan Coil Selection

The chiller produces chilled water at plant level. That chilled water must be delivered to each room or zone through cooling terminal devices. The two most common are **Fan Coil Units (FCUs)** for room-level control and **Air Handling Units (AHUs)** for zone or floor-level conditioning. Selection of the correct terminal determines the quality of comfort control, noise levels, energy consumption, and maintenance complexity.

Terminal Type	How It Works	Typical Application	Key Advantage	Key Limitation
Fan Coil Unit (FCU)	Room-mounted fan draws room air through a chilled-water coil. Individual room thermostat controls fan speed and/or water valve.	Hotel rooms, offices, apartments — anywhere individual room control is required	Very good room-level control; flexible layout; quiet at low fan speed	Each unit needs water connection, drain, power, and controls — high installation cost for many rooms

Air Handling Unit (AHU)	Centralised unit handles fresh air, filtration, heating, cooling, and humidification. Supplies conditioned air to multiple rooms via ductwork.	Open-plan offices, hospitals, schools, shopping centres — large open zones	Handles ventilation and cooling together; high filtration possible (HEPA); centralised maintenance	High space requirement for AHU and ductwork; individual room control only with VAV or reheat
Cassette FCU (ceiling)	Same as FCU but ceiling-mounted with 4-way discharge — serves larger floor area per unit	Open offices, restaurants, retail — medium-size open spaces	Covers large area per unit; aesthetically integrated in ceiling	Requires ceiling void for installation; condensate pump needed if no fall available
Chilled Beam (passive)	Chilled water coil suspended in ceiling — room air convects through coil without fan	High-end offices, classrooms, museums — where silence and air quality are priorities	Silent; no moving parts; long life; superior air quality without high air velocities	Higher capital cost; strict dew point control required to prevent condensation on beam
Variable Air Volume (VAV)	AHU supplies constant-temperature air; VAV boxes at zone level vary airflow to match load	Large multi-zone commercial buildings with varying internal loads by zone	Good energy efficiency at part load; single AHU serves many zones; flexible for office fit-outs	Complex controls; minimum airflow required for ventilation at low loads; reheat energy penalty

Worked Example: Chiller Plant Selection for a 5-Storey Office

A 5-storey office building in Addis Ababa has a peak cooling load of **620 kW** calculated by a full ASHRAE load analysis. The design team must select the chiller plant, distribution strategy, and terminal units. The building operates 12 hours per day, 5 days per week. Electricity cost: 1.5 ETB/kWh (approx. $0.027/kWh).

#	Decision — Analysis	Result / Selection
1	Chiller type: water vs air-cooled Building is 620 kW — above 500 kW threshold. Space available on roof for cooling tower. Water is available.	Water-cooled chiller — better COP, lower running cost at this scale
2	Number of chillers: 1 × 620 kW or 2 × 310 kW Redundancy required (hospital-grade building): N+1 rule applies.	2 × 400 kW chillers (one standby, one duty at full load or both at 77.5%)

3	Chiller efficiency specification ASHRAE 90.1 minimum for water-cooled centrifugal: COP 5.55. Specify above minimum.	COP ≥ 6.2 full load, IPLV ≥ 7.0 — reduces annual electricity bill significantly
4	Distribution system: primary-secondary 620 kW, 2 chillers, multiple floors — variable demand by floor and time of day.	Primary-secondary with VSDs on secondary pumps — saves ~30% pump energy
5	Chilled water temperatures Standard 6/12°C. Consider high-delta-T 6/14°C to reduce pump flow rates.	6°C supply / 14°C return (ΔT = 8°C) — reduces secondary pump flow by 25%
6	Terminal units Open-plan floors: VAV system for zone control. Individual offices at perimeter: FCUs for room control.	Central AHU + VAV for open areas; perimeter FCUs for individual offices
7	Annual energy cost estimate Average load 310 kW (50%), COP 7.0 at part load, 12h/day × 250 days/yr.	Energy = (310/7.0) × 12 × 250 = 132,857 kWh/yr ≈ 199,286 ETB/yr (~$3,586/yr)

DESIGN OUTCOME — The 2 × 400 kW water-cooled chiller plant with primary-secondary distribution, VSDs on secondary pumps, and high delta-T (8°C) chilled water design provides full redundancy, excellent part-load efficiency, and approximately 35% lower annual energy cost compared to a single air-cooled chiller of the same capacity at standard design conditions. The additional capital cost of the cooling tower and secondary pump VSDs typically pays back in 3–5 years.

Key Takeaways — MEP-006 Cooling Systems and Chillers

• **All mechanical cooling** uses the vapour compression cycle: evaporator absorbs heat, compressor raises pressure, condenser rejects heat, expansion valve drops pressure.

• **Choose water-cooled chillers** for capacity above 500 kW — COP 5–7 vs air-cooled 2.5–3.8. The cooling tower investment pays back within 3–5 years.

• **Specify IPLV not just full-load COP** — buildings rarely run at 100% load. IPLV determines actual energy bills.

• **High delta-T chilled water design** (8–10°C vs standard 6°C) reduces pump flow, pipe sizes, and energy by 20–35%.

• **Cooling tower Legionella risk** is a legal obligation, not a recommendation. Always include a water treatment specification and handover to FM team with a written scheme of control.

• **Match terminal units to the application** — FCUs for individual room control, AHU+VAV for large open zones, chilled beams where silence is critical.

• **Next lesson — MEP-007:** Ventilation and Indoor Air Quality — fresh air rates, filtration standards, heat recovery, and mechanical ventilation design.